Predicting Spotify Track Commercial Success using Random Forest

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Contents

data:https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset

1 Package load

```
library(caret)
library(tidyverse)
library(doParallel)
library(knitr)
library(rsample)
library(vip)
library(pROC)
library(kableExtra)
```

2 Data load

```r
df <- read_csv("dataset.csv")
```

3 Enabling parallel processing

```r
n_cores <- detectCores()
cl <- makePSOCKcluster(n_cores)
registerDoParallel(cl)
```

4 Dataset breakdown

```r
summary(df)
```

```
##       ...1          track_id           artists           album_name       
##  Min.   :     0   Length:114000      Length:114000      Length:114000     
##  1st Qu.: 28500   Class :character   Class :character   Class :character  
##  Median : 57000   Mode  :character   Mode  :character   Mode  :character  
##  Mean   : 57000                                                           
##  3rd Qu.: 85499                                                           
##  Max.   :113999                                                           
##   track_name          popularity      duration_ms       explicit      
##  Length:114000      Min.   :  0.00   Min.   :      0   Mode :logical  
##  Class :character   1st Qu.: 17.00   1st Qu.: 174066   FALSE:104253   
##  Mode  :character   Median : 35.00   Median : 212906   TRUE :9747     
##                     Mean   : 33.24   Mean   : 228029                  
##                     3rd Qu.: 50.00   3rd Qu.: 261506                  
##                     Max.   :100.00   Max.   :5237295                  
##   danceability        energy              key            loudness      
##  Min.   :0.0000   Min.   :0.0000   Min.   : 0.000   Min.   :-49.531  
##  1st Qu.:0.4560   1st Qu.:0.4720   1st Qu.: 2.000   1st Qu.:-10.013  
##  Median :0.5800   Median :0.6850   Median : 5.000   Median : -7.004  
##  Mean   :0.5668   Mean   :0.6414   Mean   : 5.309   Mean   : -8.259  
##  3rd Qu.:0.6950   3rd Qu.:0.8540   3rd Qu.: 8.000   3rd Qu.: -5.003  
##  Max.   :0.9850   Max.   :1.0000   Max.   :11.000   Max.   :  4.532  
##       mode         speechiness       acousticness    instrumentalness 
##  Min.   :0.0000   Min.   :0.00000   Min.   :0.0000   Min.   :0.00e+00  
##  1st Qu.:0.0000   1st Qu.:0.03590   1st Qu.:0.0169   1st Qu.:0.00e+00  
##  Median :1.0000   Median :0.04890   Median :0.1690   Median :4.16e-05  
##  Mean   :0.6376   Mean   :0.08465   Mean   :0.3149   Mean   :1.56e-01  
##  3rd Qu.:1.0000   3rd Qu.:0.08450   3rd Qu.:0.5980   3rd Qu.:4.90e-02  
##  Max.   :1.0000   Max.   :0.96500   Max.   :0.9960   Max.   :1.00e+00  
##     liveness          valence           tempo        time_signature 
##  Min.   :0.0000   Min.   :0.0000   Min.   :  0.00   Min.   :0.000  
##  1st Qu.:0.0980   1st Qu.:0.2600   1st Qu.: 99.22   1st Qu.:4.000  
##  Median :0.1320   Median :0.4640   Median :122.02   Median :4.000  
```

```
## Mean   :0.2136   Mean   :0.4741   Mean   :122.15   Mean   :3.904
## 3rd Qu.:0.2730   3rd Qu.:0.6830   3rd Qu.:140.07   3rd Qu.:4.000
## Max.   :1.0000   Max.   :0.9950   Max.   :243.37   Max.   :5.000
## track_genre
## Length:114000
## Class :character
## Mode  :character
##
##
##
```

```r
spec(df)
```

```
## cols(
##   ...1 = col_double(),
##   track_id = col_character(),
##   artists = col_character(),
##   album_name = col_character(),
##   track_name = col_character(),
##   popularity = col_double(),
##   duration_ms = col_double(),
##   explicit = col_logical(),
##   danceability = col_double(),
##   energy = col_double(),
##   key = col_double(),
##   loudness = col_double(),
##   mode = col_double(),
##   speechiness = col_double(),
##   acousticness = col_double(),
##   instrumentalness = col_double(),
##   liveness = col_double(),
##   valence = col_double(),
##   tempo = col_double(),
##   time_signature = col_double(),
##   track_genre = col_character()
## )
```

```r
df %>%
  is.na() %>%
  colSums()
```

```
##              ...1         track_id          artists       album_name
##                 0                0                1                1
##        track_name       popularity      duration_ms         explicit
##                 1                0                0                0
##      danceability           energy              key         loudness
##                 0                0                0                0
##              mode      speechiness     acousticness instrumentalness
##                 0                0                0                0
##          liveness          valence            tempo   time_signature
##                 0                0                0                0
##       track_genre
##                 0
```

5 Popularity threshold

A threshold of 50 was adopted, defining a 'Hit' as a track in the upper quartile of popularity (Top 25%).

```
threshold <- 50
```

6 Deleting unnecessary variables and transformation

```
df <- df %>%
  select(popularity,
         danceability,
         energy,
         loudness,
         speechiness,
         acousticness,
         instrumentalness,
         liveness,
         valence,
         tempo,
         duration_ms,
         explicit,
         mode,
         key,
         time_signature) %>%
  mutate(is_popular=ifelse(
    popularity>threshold,
    "Yes",
    "No"
  )) %>%
  select(-popularity) %>%
  mutate(
    is_popular=as.factor(is_popular),
    explicit=as.factor(explicit)
)
```

7 Train and test split

```
set.seed(42)
train_test_split <- initial_split(
  data = df,
  prop = 0.7
)
df_train <- training(train_test_split)
df_test <- testing(train_test_split)
```

8 Model training

```r
ctrl <- trainControl(
  method = "cv",
  number = 5,
  classProbs = TRUE,
  summaryFunction = twoClassSummary
)

model <- train(
  form = is_popular~.,
  data = df_train,
  method = "rf",
  trControl = ctrl,
  metric = "ROC",
  ntree = 200
)
```

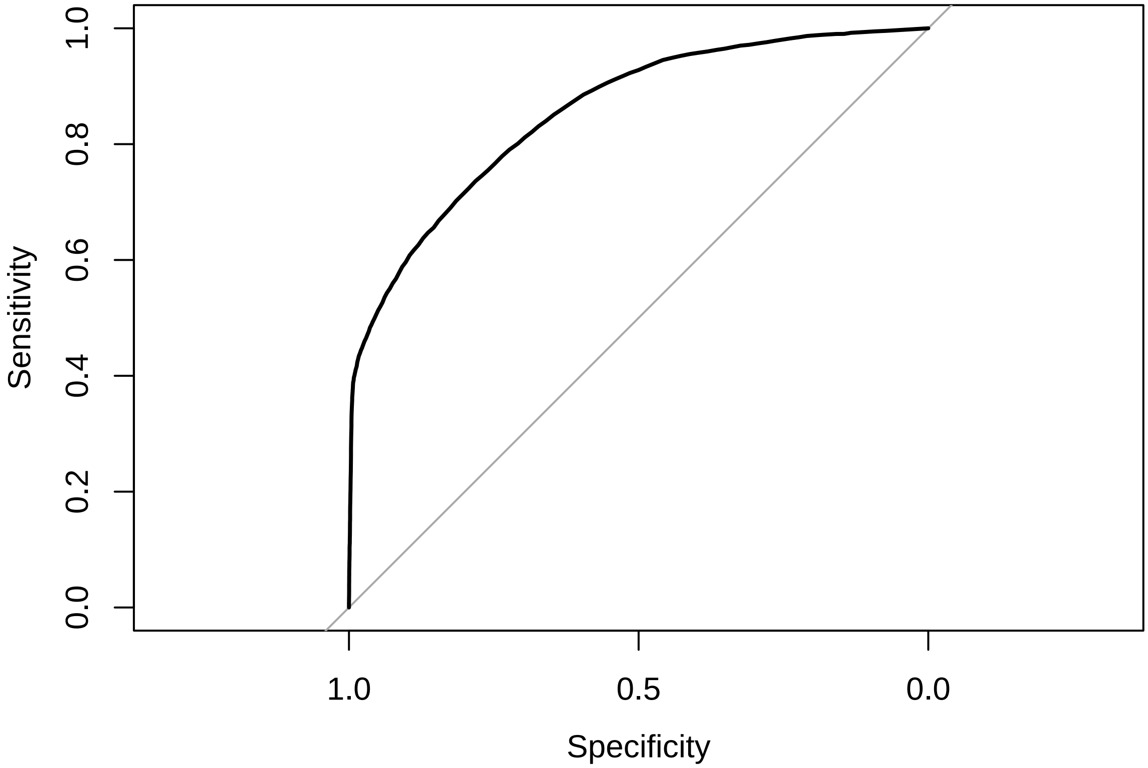
9 Predicition

```r
prob <- predict(
  object = model,
  newdata = df_test,
  type = "prob"
)[,2]

treshold_rf <- 0.5

pred <- ifelse(
  prob>treshold_rf,
  "Yes",
  "No"
)%>% as.factor()

roc_rf <- roc(
  response = df_test$is_popular,
  predictor = prob
)
plot(roc_rf)
```



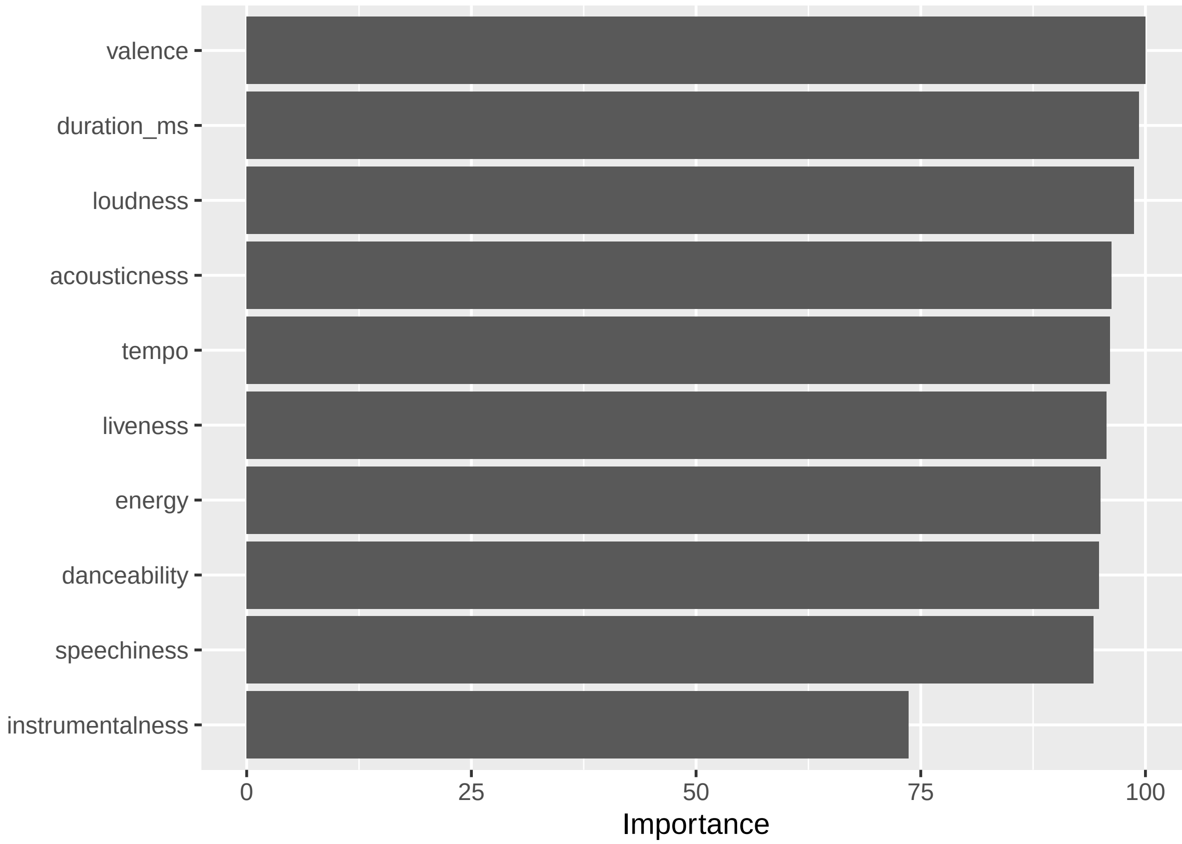

```r
roc_rf$auc
```

```
## Area under the curve: 0.8552
```

```r
vi(model)
```

```
## # A tibble: 14 x 2
##    Variable         Importance
##    <chr>                 <dbl>
##  1 valence              100
##  2 duration_ms           99.3
##  3 loudness              98.7
##  4 acousticness          96.2
##  5 tempo                 96.1
##  6 liveness              95.7
##  7 energy                95.0
##  8 danceability          94.9
##  9 speechiness           94.2
## 10 instrumentalness      73.6
## 11 key                   49.2
## 12 mode                   4.18
## 13 time_signature         1.72
## 14 explicitTRUE           0
```

```r
vip(model)
```



The model achieved an AUC score of 0.8552. This indicates that the classifier has a high discriminative ability.

The model correctly identifies most popular tracks, confirming the hypothesis that commercial success largely depends on measurable technical parameters of the track, rather than just random factors.

10 Hit profile

The 5 most influential parameters were selected.

```r
hit_profile <- df %>%
  filter(is_popular=="Yes") %>%
  summarise(
    mean_valence=mean(valence),
    mean_duration=mean(duration_ms)/60000, # to display in minutes
    mean_loudness=mean(loudness),
    mean_acousticness=mean(acousticness),
    mean_tempo=mean(tempo)
)

hit_profile %>%
```

```
  kbl(
    caption = "Hit Profile",
    booktabs = T
  ) %>%
  kable_styling(
    latex_options = c(
      "striped",
      "hold_position"),
    position = "center"
)
```

Table 1: Hit Profile

mean_valence	mean_duration	mean_loudness	mean_acousticness	mean_tempo
0.4556829	3.671402	-7.955099	0.2941638	121.4526

The HIT profile reveals that tracks more prone to becoming popular tend to convey sadness or anger (lower valence), last slightly over 3.5 minutes, have a loudness of just under -8 dB, are generally not acoustic, and maintain a tempo of around 121 BPM.